

The role of L1-dialect in L2 production

Acoustic measures on Xining and Leshan dialect speakers' production of English vowels

Ying Li

Yangzhou University

This study compared the productions of American English (AE) vowels /i/, /u/, /ε/, /ɔ/, /ɪ/, /æ/, /ʊ/, /ʌ/, and /ɑ/ between speakers of two Chinese dialects, the Leshan dialect (LSd) and the Xining dialect (XNd). The participants were native speakers of XNd ($n = 120$) and LSd ($n = 120$) with advanced English proficiency. Their vowel productions were compared with those of 22 native AE speakers. The results showed that (1) XNd and LSd participants significantly differed in F1 and/or F2 from AE participants when producing /ɪ/, /æ/, /ʌ/, and /ʊ/ and /ɪ/, /ɑ/, /ɔ/, and /ʌ/, respectively; (2) compared with AE participants, XNd participants had significantly longer durations of /ɪ/, /ʊ/, /ʌ/, /ε/, and /æ/, and LSd participants had significantly longer durations of /ɪ/, /ʌ/, and /ε/. Overall, the vowels that differed in production from that of native speakers were absent in both the L1 dialects and L1 of participants.

Keywords: L1 dialect, L2 production, vowels, Xining dialect, Leshan dialect

1. Introduction

Misproduction of segmentals can cause difficulties in intelligibility and/or comprehensibility in L2 communication (Levis, 2018; Speake, Stackhouse, & Pascoe, 2012). Predicting and detecting potential pronunciation challenges can help teachers better tailor their teaching materials (i e., focusing on the teaching and/or practicing of the potentially problematic L2 sounds). Although learners' language backgrounds have been shown to interfere with L2 sound learning (Flege, 1995, 2003; Best & Tyler, 2007), most prior research focused on L1's impact on L2 sound learning (Li & Somlak, 2019; Pittam & Ingram, 1992), with little attention being paid to L1 dialects' potential impact on L2 segmental production.

In some communities, individuals are exposed to their L1 dialect earlier and use it more frequently in daily life than their official L1. For example, in China, even though Mandarin is the official language, people in different geographical areas speak local Chinese dialects in addition to Mandarin. Chinese children are usually exposed to their local dialects prior to Mandarin — before entering kindergartens and/or primary schools where Mandarin is used for educational purposes, especially in families where both parents and grandparents¹ are from the same local dialect background.

Previous studies on Chinese speakers learning an L2 have largely ignored the impact of dialects and grouped participants as Chinese speakers in general (e.g., Chen, 2006; Chen et al., 2001; Deterding, 2006; Zhao et al., 2014). This study addresses this gap by focusing on two rarely studied and relatively unintelligible² Mandarin dialects: the Xining dialect (XNd) and Leshan dialect (LSd) (Li, 2017; Liu, 2006) as the target L1 dialects, and examined XNd and LSd speakers' production of L2-English vowels.

2. Literature review

When born into a dialect-speaking family, language learners may be exposed to the local dialect(s) more frequently and earlier than the government's official language, especially before starting formal education. Among the few available studies examining L1 dialectal influence on L2 sound learning, it was found that L1 dialects affect L2 speech perception and/or production similarly to how L1 influences L2 learning (Broselow, 1983; Gardner, 2010; Gast, 1990; Grosse & Hameyer, 1979; James, 1997; O'Brien & Smith, 2010; Seubsunk, 2000; Teasdale, 1997). For example, both Gast (1990) and James (1977) found phonetic variances between Swabian and Standard German (such as non-existence of voiced-voiceless contrasts in Swabian) led to unique errors in English segmental production (voiceless and voiced stops) by Swabian speakers. O'Brien and Smith (2010) examined English and German vowels produced by 72 English learners of German from three distinct dialectal regions. The formant values of the produced vowels varied across the participants as a function of dialectal differences. Seubsunk (2000)

1. In China, it is common for grandparents to take care of their grandchildren due to young couples' work commitments and traditional family views that see this as a duty, especially for the husband's parents.

2. XNd and LSd lack written systems, and their spoken forms are so distinct that speakers of other dialects often struggle to understand them. This underscores the importance of dialectal variation in L2 sound learning and comprehension.

investigated the production of English stop-liquid and stop-glide clusters by Thai speakers of two dialect regions differing in whether liquids are produced or deleted. Speakers of the dialect in which liquids are retained outperformed those whose dialect deletes liquids in Thai clusters.

Chinese speakers also show dialectal variations in English production. Ho (2003) found central Chinese speakers substituted /n/ for /l/, while northern Chinese speakers (particularly Beijing speakers) epenthesised /r/ in words containing /r/ or /l/. Cantonese and Fujian speakers (in southeastern China) were revealed to realize /ŋ/ as /n/ in a word-final position (Kenworthy, 1998). Deterding (2006) examined the production of English sounds by 13 adult Chinese speakers from northeastern, eastern and central China with a read-aloud task and an interview. The speakers from Liaoning (northeastern China) and Zhejiang (southeastern China) replaced /ð/ with /d/, those from elsewhere substituted /z/ for /ð/, and those from the northeastern cities of Liaoning and Shandong predominantly replaced /ʒ/ with /ɹ/.

The present study adopted theoretical bases from studies on L1 influence on L2 learning due to the lack of specific theories on L1 dialect influence. Flege's speech learning model (SLM) addresses how L2 learners expand their repertoire of phonetic categories by mapping an L2 sound to their existing L1 sound system. SLM posits that adult L2 learners initially tend to perceive unfamiliar L2 sounds through the lens of their L1 categories; when an L2 sound closely resembles an L1 sound, an "equivalence classification" occurs, often leading to difficulties in accurate L2 production (Flege et al., 1995; Flege, 2003). For example, Japanese learners of English are frequently found to merge English /ɹ/ and /l/ into the Japanese liquid /r/, particularly in early English learning stages (Bradlow, 2008; Flege, Takagi, & Mann, 1995, 1996; Idemaru et al., 2012; Lambacher, 1999). Similarly, it was found that many Cantonese speakers have difficulties distinguishing Mandarin alveolars ([tɕ], [tɕʰ], and [ɕ]) from retroflex affricates ([ʈʂ], [ʈʂʰ], and [ʂ]) and distinguishing aspirated alveolars from retroflex affricates, which are attributed to the lack of retroflexes in Cantonese (Zhang et al., 2014); English speakers tend to realize Chinese alveolar-palatals ([tɕ], [tɕʰ], [ɕ]) and retroflexes ([ʈʂ], [ʈʂʰ], [ʂ]) as postalveolar and alveolar fricatives respectively (Tutatchikoya, 1995).

Nevertheless, this does not mean a new L2 sound category cannot be established. SLM posits that with increase L2 exposure and experience, the learners are able to develop a distinct new categorial representation that leads to more accurate production of the L2 sound (Flege et al., 1995; Flege, 2003). Supporting evidence of this view comes from a number of studies using length of residence as a variable to measure L2 experience, showing that extended exposure to L2-speaking environment enhances learners' performance in both L2 segments and prosody (Flege, Bohn, & Jang, 1997; Trofimovich & Baker, 2006). Studies

on phonetic training demonstrating improved perception and production with intensive exposure provide further evidence in support of the view that increased L2 input and experience can lead to the establishment of new L2 sound categories (Aliaga-García & Mora, 2009; Iverson & Evans, 2009; Li & Somlak, 2019).

However, counterevidence is found in other studies where prolonged residence in an L2-speaking community does not always eliminate a foreign accent (Bannert, 1984; Thompson, 1991). For example, Flege, Munro, & Skelton (1995) found that increased length of residence in the US did not improve the intelligibility of English word-final stops produced by Mandarin and Spanish speakers. Likewise, the Spanish-Catalan bilinguals in Pallier, Bosch, & Sebastián-Gallés (1997), who had at least 12 years of exposure to Catalan and had extensive use of Catalan during the exposure, had not fully mastered the perceptual distinction between Catalan /e/ and /ɛ/.

Based on the SLM, McAllister, Flege, & Piske (2002) developed the Feature Hypothesis, suggesting that a mismatch of phonetic features used to signal contrast in an L1 with L2 phonetic categories may block L2 phonetic category formation. Specifically, L2 features not used for contrast in the L1 pose challenges for L2 phonetic category formation (McAllister et al., 2002). For example, McAllister et al. (2002) tested 20 L1-English, L1-Spanish, and L1-Estonian speakers' perception and production of Swedish. The results showed that L1-Estonian performance was very similar to that of the native Swedish participants, and L1-English participants' performance was more similar to native Swedish speakers than that of L1-Spanish participants. These findings were explained by the duration contrast in the participants' L1 — duration is a prominent feature in Swedish and Estonian but not in English and Spanish, whereas native English speakers make greater use of duration to signal contrast than native Spanish speakers (McAllister et al., 2002).

In addition, Best & Tyler's (2007) Perceptual Assimilation Model-L2 (PAM-L2) proposes that L2 learners assimilate new sounds into their L1 phonetic categories based on articulatory similarity. The assimilation results are classified into four types depending on L1-L2 articulatory features: (1) only one phonetic category is permanently assimilated as an equivalent L1 sound; (2) two L2 sounds are assimilated to a single L1 sound, with one sound being perceived as a better exemplar than the other; (3) two L2 sounds are perceived as a single L1 sound, with an equal degree of goodness; and (4) no assimilation occurs, and new phonetic categories need to be established to perceive the L2 sounds (Best & Tyler, 2007). In line with the "equivalence classification" proposed in the SLM, condition (3) is predicted to be the most challenging for L2 sound learning, such as Japanese speakers' assimilation of English /ɹ/-/l/ into a single Japanese category /R/.

The PAM-L2 and SLM both agree that greater similarity between L1 and L2 sounds makes L2 acquisition more difficult, but with more experience, new phonetic categories can be established. The revised version of the SLM (SLM-r) expands this view, suggesting that although the neural mechanism of learning a new language remains intact throughout one's life, the successful L2 category formation depends on (1) the precision of established L1 categories when L2 learning begins; (2) the perceived phonetic distance between an L2 sound and its L1 counterpart; and (3) the quantity and quality of L2 input (Flege & Bohn, 2021). For instance, increased English experience helps Japanese speakers better distinguish /ɹ/ -/R/ than /l/-/R/; Japanese speakers were predicted to continue to perceive and produce /ɹ/ more accurately than /l/, regardless of the amount of English input (Flege & Bohn, 2020).

Despite ongoing research, the extent of L1 influence on L2 sound learning remains unclear. This study aims to shed further light on this issue from the perspective of L1 dialectal influences on L2 sound learning, focusing on the production of English vowels by speakers of two different Chinese dialects. These vowels were chosen because: (1) Prior studies noted common vowel misproduction among Chinese speakers (e.g., /ɹ/, /æ/, and /ʊ/) but overlooked dialect backgrounds (Chen, 2006; Ho, 2003). (2) Compared with consonants, vowel articulation involves subtle tongue position changes that alter vocal tract resonance, making it challenging for L2 learners (Ball & Gibbon, 2013). (3) Among the few studies that evaluated L1 dialectal influences on L2 segmental learning, most of them assessed participants' production with native speakers' subjective assessment rather than acoustical analysis (e.g., Deterding, 2006; Deterding & Poedjosoedarmo, 1998). This study acoustically analyzes and compares the English vowel production of dialect speakers to native English speakers.

3. The vowel systems of XNd, LSd, Mandarin and Midland American English

The target dialects were two rarely studied ones from Mainland China, the XNd and LSd. Chinese consists of minority dialects (e.g., Tibetan dialect groups) and various Mandarin dialects, such as Wu, Gan, Xiang (Coblin, 2000; Escure, 1997). Both XNd and LSd are classified as Mandarin dialects (Norman, 2003).

XNd is one of the three main dialects spoken in Qinghai Province, located in Northwest China. It is used by approximately 2.5 million people in Xining, the provincial capital (see Figure 1) (Dede, 2007; Zhang, 1984). Research on XNd has identified nine basic vowels: /i/, /a/, /u/, /y/, /ɛ/, /ɔ/, /ā/, /ǎ/, and /ǐ/ (Dede, 2007; Zhang, 1984).



Figure 1. Map showing the locations of Xining and Leshan

The second dialect in this study is LSd, spoken by approximately 3.3 million people in Leshan city, located in southeastern Sichuan Province (see Figure 1). LSd has a phonetic system comprising seven vowels: /i/, /æ/, /u/, /ʊ/, /y/, /ə/, and /e/ (Lai, 2000).

Mandarin (Standard Chinese/Putonghua) is the official language of the People’s Republic of China, used for education, media, formal situations, and other official transactions (Norman, 1998). Like other dialect speakers, XNd and LSd speakers learn Mandarin after acquiring their local dialects. Mandarin is generally agreed to have six basic vowels: /i/, /a/, /u/, /ɤ/, /ɿ/, and /ə/ (Barkana & Patel, 2020; Dede, 1999; Zee, 2001).

English has a comparatively larger vowel inventory that varies across English-speaking regions. Participants in the present study were primarily exposed to American English, particularly through school materials, TV series, and movies. The native English-speaking participants had a Midland American accent, characterized by eleven vowels /i/, /u/, /ɛ/, /ɔ/, /ɪ/, /ei/, /æ/, /ʊ/, /oʊ/, /ʌ/, and /ɑ/ (Labov, Ash, & Boberg, 2008). This study focused on the nine monophthongs /i/, /u/, /ɛ/, /ɔ/, /ɪ/, /æ/, /ʊ/, /ʌ/, and /ɑ/.

On the whole, as displayed in Figure 2, the common vowels among XNd, LSd and the target (AE) are /i/ and /u/. /i/, /u/, /ɔ/, and /ɛ/ exist in both XNd and AE; /i/, /u/, /æ/, and /ʊ/ occur in both LSd and AE; /y/ exists in both XNd and LSd but not in AE; and the remaining vowels (XNd: /a/, /ã/, /ɜ̃/, and /ẽ/; LSd: /e/ and /ə/; AE: /ɪ/, /ʌ/, and /ɑ/) do not exist in the other phonetic inventories.

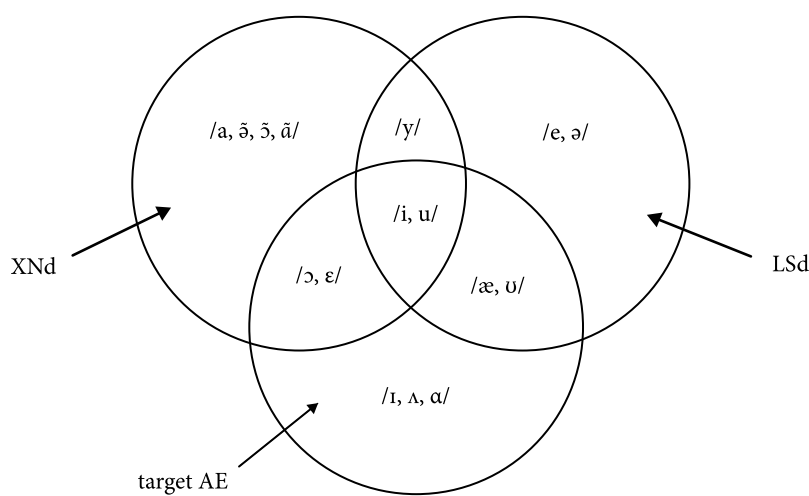


Figure 2. Vowels in XNd, LSd and target AE

Considering the influence of Mandarin, XNd and LSd participants may have command of Mandarin vowels (/a/, /ɤ/, and /ə/) in addition to those of their own dialect (see Figure 3). With the addition of these vowels, the common vowels between XNd and AE remain /i/, /u/, /ɔ/, and /ɛ/, and the common vowels between LSd and AE remain /i/, /u/, /æ/, and /ʊ/. Nevertheless, as discussed by Flege (1995, 2003), the similarities/dissimilarities of vowels across languages cannot be compared based only on IPA. Therefore, native AE speakers were included as a control group in this study to compare the participants' production to the target AE vowels.

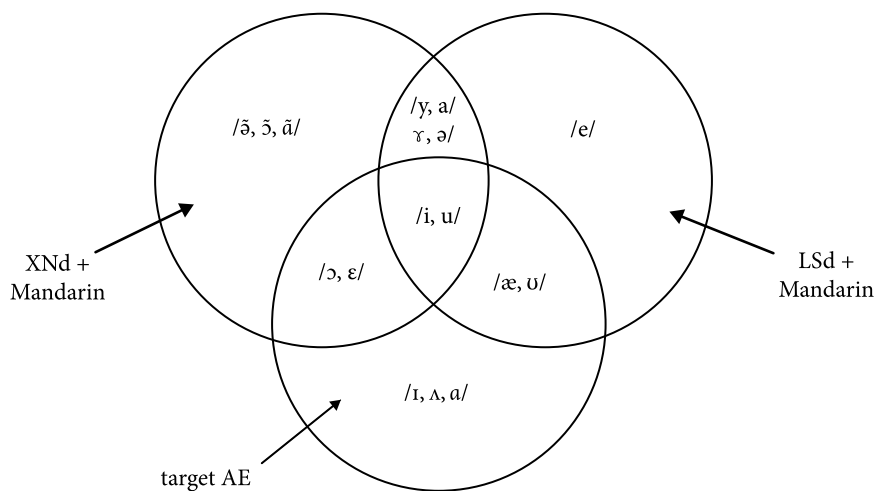


Figure 3. Vowels in XNd + Mandarin, LSd + Mandarin, and target AE

4. Research question and hypotheses

4.1 Research question

To what extent, if any, are the the XNd and LSd participants’ production of target AE vowels acoustically similar to that of native AE speakers and each other’s dialects in terms of F1, F2 and duration?

4.2 Hypotheses

Based on SLM and PAM-L2, which suggest that greater dissimilarity between L2 and L1 sounds facilitates L2 learning – and previous studies on Chinese speakers’ English vowel production (e.g., Chen, 2006; Ho, 2003), the following predictions were made:

- 1. XNd participants are expected to have difficulties in distinguishing /i/-/ɪ/, /u/-/ʊ/, /a/-/æ/, and /ʌ/-/ɑ/; they might also nasalize the AE vowels /ɑ/ and /ɔ/, depending on whether they can distinguish /ǎ/-/ɑ/ and /ɤ/-/ɔ/.
- 2. LSd participants were expected to have challenges in distinguishing /i/-/ɪ/ and /ʌ/-/ɑ/.
- 3. Drawing from the author’s English learning and teaching experience in China, it is common for Chinese students to use only durational difference to distinguish lax – tense vowels (/ɪ/-/i/ and /ʊ/-/u/), often due to early misguidance from Chinese teacher. Thus, it was hypothesized that both XNd and LSd participants might struggle to distinguish /ɪ/-/i/ and /ʊ/-/u/.

5. Participants

5.1 Participants

Native XNd, LSd and Midland American English speakers were recruited with online advertisements on university and local job websites, as well as Facebook. All participants reported no speaking, hearing, or learning difficulties. They volunteered for the study and provided written informed consent, with the assurance that their recordings would be destroyed after the study's completion.

XNd participants were recruited with the following inclusion criteria: (1) born and raised in Xining, Qinghai;³ (2) not fluent in another Chinese dialects or languages other than English, thus ruling out the influence of another language/dialect on their production performance; (3) began learning English as an L2 at the age of 6–9⁴ years; (4) had passed the CET-6⁵ and scored between 550 and 600 (advanced English proficiency), hence ruling out a potential bias due to differences in L2 proficiency levels on individuals' pronunciation accuracy. As a result, 120 students aged 19–27 years (60 females, 60 males; $M_{age} = 23.6$ years) were selected from three universities in Xining, Qinghai.

LSd participants were recruited with the same criteria as those of XNd participants, except they must be born and raised in Leshan, Sichuan Province. A total of 120 university students aged 18–27 years (62 females, 58 males; $M_{age} = 22.8$ years) were recruited from universities in Leshan, Sichuan.

The native AE participants were 22 college students (11 males, 11 females; aged 19–28 years; $M_{age} = 23.24$ years) from a university in Columbus, US. They were born and raised in Omaha (Nebraska) or Columbus (Ohio), which gave them Midland English accents (Jacewicz, Fox, & Salmons, 2011; Labov et al., 2008). All of them reported learning Spanish as an L2 at a lower-intermediate level and had no knowledge of XNd, LSd, Mandarin or any other Chinese dialects.

3. Dialects in Qinghai province are classified into XNd (Huangshui Chinese, Dede (2007)), Minhe and Ledu dialect groups (Zhang, 1984). All XNd participants were born in Xining to ensure that their primary dialect was XNd, not another Qinghai dialect.

4. The selected age range aligns with the local educational system in mainland China, where children typically start primary school at ages 6–7. Formal English courses are offered in Grades 1–6 across various schools.

5. The CET is a national English proficiency test for college students in China, available to all non-English major students. It consists of two levels of test: CET-4 and CET-6 (maximum score: 710). Scores between 550 and 600 in CET-6 are equivalent to a 7 on the IELTS test (Xiang-dong, 2007), which corresponds to a C1 level on the CEFR scale. Additionally, all XNd and LSd participants had received or were receiving university education in China, implying years of instruction in and use of Mandarin.

5.2 Materials

Following prior studies (Hillenbrand et al., 1995; Peterson & Barney, 1952), the nine AE monophthong vowels /i/, /u/, /ɛ/, /ɔ/, /ɪ/, /æ/, /ʊ/, /ʌ/, and /ɑ/ were embedded in /hVd/ words (e.g., *head*, *hid*, *head*, *had*, *hod*, *hawed*, *hood*, *who'd*, and *hud*) to maintain vowel clarity and minimize coarticulation effects (O'Brien & Smith, 2010; Peterson & Barney, 1952). A pilot study with 10 XNd speakers revealed mispronunciations of *hod* as [həd] and *hud* as [hʌd] due to orthographic influence and unfamiliarity. To address this, *hod* and *hud* were replaced with *god* and *bud* – two words more frequently used by Chinese speakers. However, another test showed variations: *god* was produced as [gɑ:d] ($n=1$), [gad] ($n=7$) or [gʌd] ($n=2$), and *bud* was produced as [bʌd] ($n=4$) or [bad] ($n=6$) by the same 10 XNd speakers. The final 9 stimulus words were then embedded in the carrier sentence “I say the word _____”, repeated three times in random order, 27 recordings.

A further pilot study with 4 AE speakers (2 females, 2 males) confirmed the vowels in the 9 stimulus words (*head*, *hid*, *head*, *had*, *god*, *hawed*, *hood*, *who'd*, *bud*) were pronounced as /i/, /u/, /ɛ/, /ɔ/, /ɪ/, /æ/, /ʊ/, /ʌ/, and /ɑ/, respectively. The choice of /hVd/, /gVd/ and /bVd/ contexts ensured no nasal consonant affected the vowels. Therefore, if XNd speakers produced /ɑ/ and /ɔ/ with nasalization, it could be attributed to their dialect's influence, which included /ā/ and /ō/, rather than consonant assimilation.

5.3 Procedure

Recordings for XNd and LSd participants were conducted in quiet rooms at their universities. Participants read prepared sentences displayed on a laptop screen and were given 10 minutes to prepare. They read the sentences aloud clearly and reported knowing all the words. Recording was done using a high-quality audio recorder (Roland-05; 16-bit mono, 44.1 kHz), with files saved in WAV format. AE participants were recorded in a quiet room at their university by a phonetically trained native AE speaker using the same procedure as that for XNd participants.

The recordings were segmented and annotated using Praat 6.1.32 (Boersma & Weenink, 2020). The target vowels were identified from the onset to offset of voicing or upon a sudden drop in intensity. The segmentation was refined based on auditory and spectrogram analysis by the same AE speaker who recorded the AE participants. The analysis focused on the target vowels' first and second formant frequencies (F1 and F2) and durations, which are key indications in vowel quality perception (e.g., Hillenbrand et al., 1995; Jia et al., 2006; Peterson & Barney, 1952). Formant measurements at each vowel's midpoint and durations were extracted

with Praat scripts and compiled into an Excel file, resulting in 9,720 tokens for both XNd and LSd speakers, and 2,592 for AE speakers. Due to misreading, 41 tokens from XNd (0.42%) and 24 from LSd (0.24%) were excluded, leaving a total of 21,967 tokens ($2,592 + 9,720 - 41 + 9,720 - 24$).

Among the remaining tokens, outliers were identified if duration differences exceeded 3–11 ms or formant differences exceeded 44 Hz across repetitions (Hillenbrand et al., 1995; Clopper et al., 2005). As a result, 51 outliers (3 from AE participants; 26 from XNd participants; 22 from LSd participants) out of 21,967 tokens (0.23%) were excluded from the data. The investigator rechecked the data and confirmed the reliability of the AE participants' assessment. The remaining tokens were then normalized with Lobanov's formulas (Lobanov, 1971) to exclude physiological differences caused by sex and/or age:

$$F_{n(V)}^N = (F_{n(V)} - MEAN_n) / S_n \quad (1)$$

$$D^N = D / MEAN_D \quad (2)$$

In Formula (1), $F_{n(V)}^N$ is the normalized formant value for formant n (F_n) of a vowel (V); $F_{n(V)}$ is the raw formant value produced by a speaker; $MEAN_n$ represents the speaker's mean F_n ; and S_n is the speaker's standard deviation of F_n . In Formula (2), D^N is the normalized duration; D represents the original duration and $MEAN_D$ is the mean duration value produced by a speaker.

To assess nasalization of /a/ and /ɔ/ in XNd productions (Hypothesis (2)), three raters (the AE individual who recorded AE participants and 2 native AE linguistics graduate students from Ohio) rated the nasal features on a 5-point Likert scale (from 0 = nonnasalized to 5 = nasalized) regardless of production accuracy.

Additionally, 20% ($n = 24$) of XNd participants' productions for /a/ and /ɔ/ were randomly selected and checked by the investigator for nasal acoustic features. Nasal vowels typically show low-frequency features such as: (1) lower amplitude and broader F_1 values than nonnasalized counterparts; (2) fixed peaks at 250 and 2000 Hz; and (3) a variable peak at 900 Hz (Delattre, 1965, 1966). Specifically, nasal /ã/ displays a more compact spectrum and bear more spectral features of a grave sound (a sound perceived as having a secondary and lesser intensity emphasis compared to an acute sound) than /a/, which also has a lower F_2 that merges with F_1 to create a large pole centered at approximately 900 Hz; in contrast, nasal /ɔ̃/ has an absent or weakened F_2 ($F_2 = 0 \sim 20$ Hz) that shifts toward higher frequencies⁶ (Delvaux et al., 2002).

6. Due to the limited research on the acoustic features of /ã, ɔ̃/ in XNd, the characteristics presented here are drawn from studies of the vowels in French.

6. Results

To address the research question, XNd and LSd participants’ vowel productions are compared with both those of native AE participants and within their own dialect groups.

6.1 F1 and F2 values

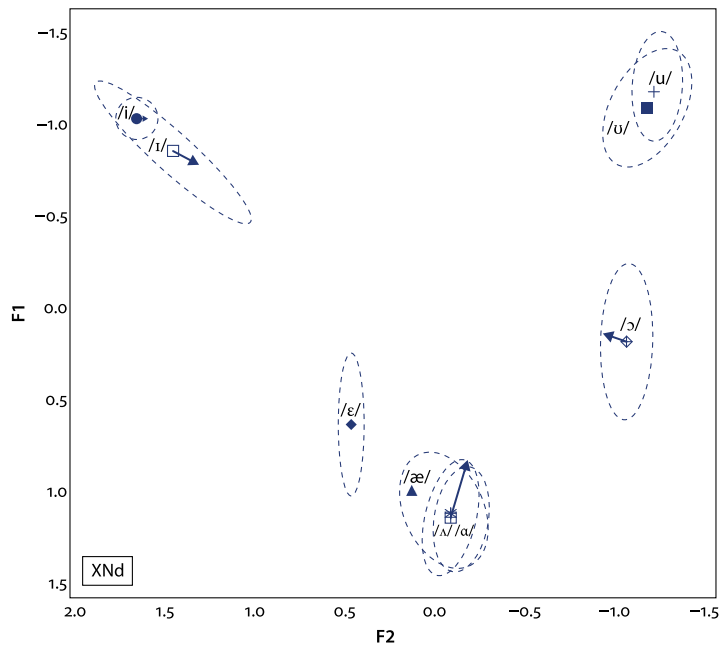


Figure 4. Mean F1 × F2 vowels Lobanow normalized – XNd participants

Figures 4, 5 and 6⁷ display the normalized mean F1 and F2 values for the vowels produced by XNd, LSd and AE participants. Overlaps occurred among XNd participants for /i/-/ɪ/, /u/-/ʊ/ and /æ/-/ʌ/-/ɑ/, and among LSd participants for /i/-/ɪ/, /ɛ/-/ɔ/ and /ʌ/-/ɑ/. In contrast, AE participants showed clear vowel space differentiation. Nevertheless, XNd productions of /ɛ/ and /ɔ/ and LSd productions of /u/, /ʊ/, and /æ/ aligned closely with AE speakers’ productions.

7. The target vowels are represented by different symbols (diamonds, squares, etc.) in Figure 4, 5, 6.

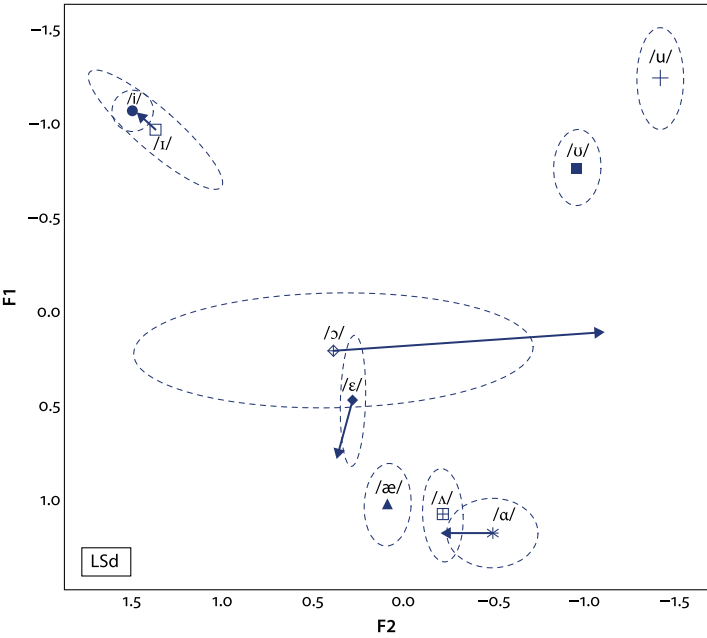


Figure 5. Mean F1 x F2 vowels Lobanov normalized – LSd participants

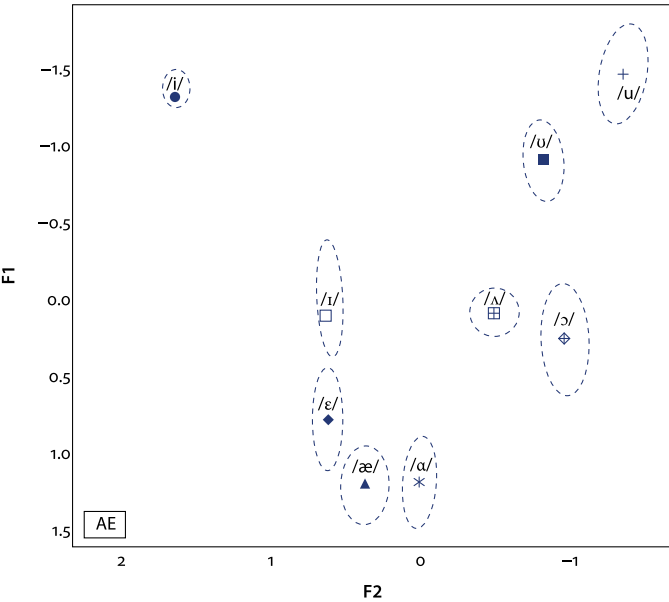


Figure 6. Mean F1 x F2 vowels Lobanov normalized – AE participants

To assess significant differences from AE vowel productions, one-way ANOVAs were conducted using individual participants' F1 and F2 values as dependent variables and their L1 dialect as the independent variable.

According to the results (see Tables 1, 2, and 3), XNd participants' F1 and F2 values for /i/, /u/, /ɔ/, /ɛ/, and /ɑ/ were not significantly different from those of AE participants ($p > 0.05$), with i/, /u/, /ɔ/, and /ɛ/ present in both XNd and AE. However, significant differences in F1 and/or F2 values were revealed in XNd participants' production of /ɪ/, /æ/, /ʊ/, and /ʌ/, which are absent in XNd and Mandarin (see Figures 2 and 3). Specifically, XNd participants displayed significantly lower F1 but higher F2 values than AE participants when producing /ɪ/ (F1: $F(1, 141) = 131.1, p < 0.001$; F2: $F(1, 141) = 107.3, p < 0.001$), indicating a higher, more anterior tongue position than AE participants. For /æ/, XNd participants displayed no significant differences in F1 ($p > 0.05$) but significantly lower F2 than AE participants ($F(1, 141) = 30.2, p < 0.001$). For /ʊ/, XNd participants showed significantly higher F1 ($F(1, 141) = 496.0, p < 0.001$) and lower F2 ($F(1, 141) = 5.1, p < 0.001$) than AE participants. When producing /ʌ/, XNd participants' F1 and F2 were both significantly higher than those of AE participants (F1: $F(1, 141) = 208.0, p < 0.001$; F2: $F(1, 141) = 129.4, p < 0.001$).

Table 1. Comparisons on AE-XNd and AE-LSd participants' F1 and F2 values in producing the target vowels

Vowel	Formant	(I) L1 dialect	(J) L1 dialect	Mean difference (I-J)	S.E.	Sig.
/i/	F1	AE	XNd	-3.1	3.2	1
			LSd	-5.8	3.6	0.32
	F2	AE	XNd	27.5	17.5	0.35
			LSd	36.7	17.1	0.11
/ɪ/	F1	AE	XNd	141.6 [*]	11.5	< 0.001
			LSd	150.2 [*]	10.9	< 0.001
	F2	AE	XNd	-652.0 [*]	61	< 0.001
			LSd	-694.2 [*]	60.4	< 0.001
/æ/	F1	AE	XNd	6.7	9	1
			LSd	1.7	8.2	1
	F2	AE	XNd	167.6 [*]	26.2	< 0.001
			LSd	-4.7	22.3	1
/ɑ/	F1	AE	XNd	-2.8	7.5	1
			LSd	-21.3 [*]	7.1	0.01
	F2	AE	XNd	9.2	29.8	1
			LSd	166.3 [*]	27.3	< 0.001

Table 1. (continued)

Vowel	Formant	(I) L1 dialect	(J) L1 dialect	Mean difference (I-J)	S.E.	Sig.
/ɔ/	F1	AE	XNd	16.3	13.4	0.53
			LSd	2.4	12	1
	F2	AE	XNd	7.2	101	1
			LSd	-973.3 [*]	115	<0.001
/ʊ/	F1	AE	XNd	-468.7 [*]	15.1	<0.001
			LSd	-1.1	16.4	1
	F2	AE	XNd	260.9 [*]	29.3	<0.001
			LSd	-5.5	25.8	1
/u/	F1	AE	XNd	0.3	9.3	1
			LSd	-2	9.7	1
	F2	AE	XNd	-10.1	20.7	1
			LSd	-7	21.4	1
/ʌ/	F1	AE	XNd	-138.6 [*]	8.9	<0.001
			LSd	-149.3 [*]	8.5	<0.001
	F2	AE	XNd	-274.3 [*]	19	<0.001
			LSd	-300.7 [*]	21.7	<0.001
/ɛ/	F1	AE	XNd	16.3	11.8	0.5
			LSd	28.3	10.5	0.05
	F2	AE	XNd	-3.8	10.1	1
			LSd	10.4	9.6	0.99

^{*} Refers to a difference was statistically significant on a 0.05 level.

Table 2. Post Hoc tests on XNd, LSd and AE participants' durational values in producing the /ɪ, ʊ, ʌ, ɛ/

Vowel	(I) L1/L1 dialect	(J) L1/L1 dialect	Mean difference (I-J) (in ms)	S.E.	Sig.
/ɪ/	AE	XNd	-55.3 [*]	5.8	<0.001
		LSd	-49.6 [*]	5.8	<0.001
/ʊ/	AE	XNd	-32.7 [*]	4.7	<0.001
		LSd	1.4	4.7	1.0
/ʌ/	AE	XNd	-44.6 [*]	6.7	<0.001
		LSd	-39.6 [*]	6.7	<0.001
/ɛ/	AE	XNd	-35.8 [*]	6.6	<0.001
		LSd	-32.3 [*]	6.6	<0.001
/æ/	AE	XNd	-54.3 [*]	6.6	<0.001
		LSd	-2.2	2.9	0.17

^{*} Refers to a difference was statistically significant on a 0.05 level.

Table 3. Means and SDs of the participants' F1 and F2 values in the production of /ɪ, æ, ɑ, ɔ, ʊ, ʌ/

Vowel	Acoustic cue	Participants	Mean	Std. deviation
/ɪ/	F1	XNd	477.2	54.4
		LSd	468.6	44.6
		AE	618.8	46.9
	F2	XNd	2649.0	292.9
		LSd	2691.2	243.0
		AE	1997.0	71.1
/æ/	F1	XNd	745.6	40.3
		LSd	750.6	30.1
		AE	752.3	31.3
	F2	XNd	1673.2	137.1
		LSd	1845.4	87.2
		AE	1840.8	93.2
/ɑ/	F1	XNd	752.6	36.8
		LSd	771.1	25.9
		AE	749.8	36.3
	F2	XNd	1620.1	87.4
		LSd	1463.0	165.8
		AE	1629.3	66.8
/ɔ/	F1	XNd	621.0	60.1
		LSd	634.9	43.0
		AE	637.2	43.9
	F2	XNd	1061.9	89.2
		LSd	2042.4	725.2
		AE	1069.1	91.5
/ʊ/	F1	XNd	965.1	97.5
		LSd	497.6	28.2
		AE	496.4	31.5
	F2	XNd	889.8	162.4
		LSd	1156.2	85.2
		AE	1150.7	79.9

Table 3. (continued)

Vowel	Acoustic cue	Participants	Mean	Std. deviation
/Λ/	F1	XNd	755.2	44.3
		LSd	757.9	34.8
		AE	616.6	18.5
	F2	XNd	1617.7	106.3
		LSd	1644.1	71.6
		AE	1343.5	89.4

LSd participants displayed no significant differences in F1 and F2 values from AE participants for /i/, /æ/, /ʊ/, /u/, and /ɛ/ ($p > 0.05$), with /i/, /æ/, /ʊ/, and /u/ present in both LSd and AE. However, significant F1 and/or F2 differences were found for /ɪ/, /ɑ/, /ɔ/, and /Λ/, which are absent in LSd and Mandarin (see Figures 2 and 3). For /ɪ/, LSd participants had significantly lower F1 but higher F2 than AE participants (F1: $F(1, 141) = 207.2, p < 0.001$; F2: $F(1, 141) = 175.8, p < 0.001$). For /ɑ/, LSd participants showed significantly higher F1 than AE participants ($F(1, 141) = 10.9, p < 0.001$) but lower F2 ($F(1, 141) = 21.4, p < 0.001$). In /ɔ/, their F1 values were not significantly different ($p > 0.05$), whereas LSd participants' F2 values were significantly higher than those of AE participants ($F(1, 141) = 39.3, p < 0.001$). Regarding /Λ/, similar to XNd participants, LSd participants showed significantly higher F1 values ($F(1, 141) = 344.1, p < 0.001$) and F2 values ($F(1, 141) = 302.8, p < 0.001$) than AE participants.

6.2 Durational results

Post hoc tests indicated significant differences in the duration of lax vowels (/ɪ/, /ʊ/, /Λ/, /ɛ/, and /æ/) among XNd, LSd and AE participants ($p < 0.05$), but no significant differences for tense vowels (/i/, /ɑ/, /ɔ/, and /u/; $p > 0.05$). Table 2 shows that both XNd (/ɪ/: Mean=108.8; /Λ/: Mean=107.9; /ɛ/: Mean=100.5) and LSd (/ɪ/: Mean=103.1; /Λ/: Mean=102.9; /ɛ/: Mean=97.0) participants produced /ɪ/, /Λ/, and /ɛ/ with significantly longer durations compared to AE participants (/ɪ/: Mean=53.6; /Λ/: Mean=63.4; /ɛ/: Mean=64.7). For /ʊ/, XNd participants (Mean=98.4) also had a significantly longer duration than AE participants (Mean=65.7), while LSd and AE participants' durations did not differ significantly ($p > 0.05$).

Overall, XNd participants produced /ɪ/, /ʊ/, /Λ/, /ɛ/, and /æ/ with significantly longer durations than AE participants, while LSd participants had significantly longer durations for /ɪ/, /Λ/, and /ɛ/ compared to AE participants.

6.3 XNd speakers’ productions of /a, ɔ/

Raters observed no nasal features in XNd participants’ productions of /a/ and /ɔ/ (Likert score=0 for all the assessed stimuli). Similarly, spectral analysis of 24 randomly selected instances XNd participants’ production of /a/ and /ɔ/ confirmed they were oral-like rather than nasal-like. Although productions of /a/ and /ɔ/ resembled /ʌ/ and /a/, respectively, none exhibited low-range frequencies (0–2500 Hz) nasal characteristics as described earlier (see *Procedure*). Figure 7 illustrates a female XNd participant’s production of /a/, displaying no nasal signs (such as a broad F1 value, or a combined F1 and F2 peak at approximately 900 Hz for /ã/; absent or weakened F2 for /ɔ̃/ (Delattre, 1965, 1966; Delvaux et al., 2002)).

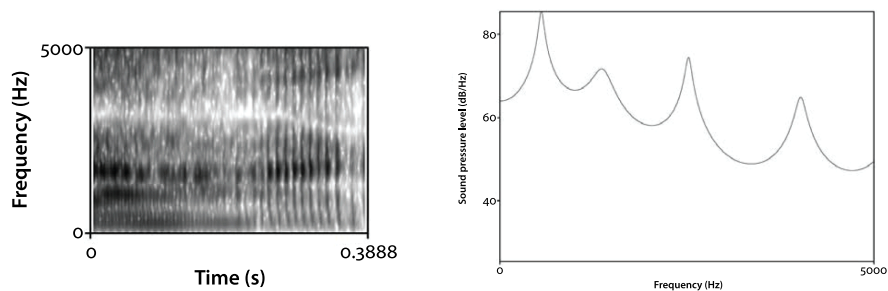


Figure 7. A sample waveform and spectral slice of /a/ produced by a female XNd participant

7. Discussion

The study explored the production of nine AE vowels by Chinese speakers of XNd and LSd, focusing on F1, F2, and duration comparisons with native AE speakers. The first finding was that significant overlaps in F1 and F2 values were found among XNd participants’ production of /i/-/ɪ/, /u/-/ʊ/, and /æ/-/ʌ/-/a/, and among LSd participants’ production of /i/-/ɪ/, /ɛ/-/ɔ/, and /ʌ/-/a/. In contrast, native AE speakers exhibited distinct vowel spaces. This suggests difficulty for XNd and LSd participants in distinguishing these contrasts.

This difficulty may stem from the absence of one or two sounds in each of these contrasts in Mandarin and their respective dialects, supporting the view that L2 sounds missing from learners’ L1 can hinder the learning of the sounds (Bradlow, 2008; Lambacher, 1999). This finding aligns with previous research where language learners struggle with the correct realization of non-native sounds lacking in their L1/L1 dialect, such as Cantonese speakers’ difficulties in distinguishing Mandarin alveolars from retroflex affricates (Zhang et al., 2014), Eng-

lish speakers' incorrect realization of Mandarin alveolar-palatals and retroflexes (Tutatchikoya, 1995), and Japanese speakers' challenges distinguishing English /ɪ/-/i/ (Flege, et al., 1995, 1996).

A closer analysis showed that, similar to previous studies where L2 speech perception/production varied by L1 dialect (e.g., Gast, 1990; James, 1977; O'Brien & Smith, 2010; Seubsunk, 2000), XNd and LSd participants differed in their production of /ʌ, ɑ, æ, ʊ, u, ɪ, i/. This finding is consistent with earlier research highlighting Chinese speakers' varied realizations of the same English vowels across dialects (Deterding, 2006; Ho, 2003; Kenworthy, 1998). Specifically, XNd participants struggled to distinguish /æ/-/ʌ/-/ɑ/ by producing similar F1/F2 values for the three sounds. In contrast, LSd participants had difficulty differentiating /ʌ/-/ɑ/, while their production of /æ/ occupied a distinct vowel space similar to that of native AE participants. This may be because the presence of /æ/ in LSd, whereas XNd and Mandarin lack all of the three vowels.

Similarly, XNd participants showed difficulty distinguishing /ʊ/-/u/, showing by overlapping vowel spaces in Figure 4, whereas LSd participants produced the two sounds with distinct F1/ F2 values (Figure 5). This might due to the presence of /ʊ/ in LSd, which lacks in XNd and Mandarin. Likewise, both XNd and LSd groups struggled to differentiate /ɪ/-/i/, which might be attributed to the non-presence of /ɪ/ in both dialects and Mandarin.

The similar performance of XNd and LSd participants in producing /ɪ/-/i/ and the difficulties faced by XNd participants in distinguishing /ʊ/-/u/ support SLM, which hypothesizes that L2 learners may have difficulties acquiring L2 sounds that are similar but not identical to an L1 counterpart; the more similar an L2 sound is to its L1 counterpart, the harder it is to form. New phonetic category (Flege et al., 1995; Flege, 2003). The overlaps shown in Figures 4 and 5 might be considered as examples of 'equivalence classification' proposed by SLM, where non-native sounds are perceived as instances of the same category as an L1 sound rather than as distinct new one. Therefore, XNd and LSd participants might have perceived non-native /ɪ/ and/or /ʊ/ as equivalents to /i/ and/or /u/, respectively, in their L1/L1 dialect.

These F1/F2 overlaps may also be explained by PAM-L2, which states that learners tend to assimilate L2 sounds into the closest articulatory-similar sound in their L1 phonetic inventory. For example, as /ɪ/ is absent in Mandarin, XNd, and LSd, participants may have assimilated it to the similar sound /i/ in Mandarin and/or their dialect. In fact, compared to AE participants, XNd and LSd speakers produced /ɪ/ with a higher and more anterior tongue position, causing it to overlap with /i/.

While XNd participants correctly produced /ɑ/, they struggled with /æ/ and /ʌ/. Significant differences in F1/F2 values were observed for /ʌ/-/ɑ/ and /æ/-/ʌ/,

but not for /æ/-/ɑ/. According to PAM-L2, XNd participants may have assimilated /æ/-/ɑ/ into one category (/ɑ/), while /ʌ/ maintained a greater phonetic distance from /ɑ/. As hypothesized by SLM-r, the perceived L1-L2 phonetic distance is crucial for establishing new sound categories (Flege & Bohn, 2021). Notably, XNd participants performed better with /æ/ than /ʌ/ — their production of /æ/ significantly differed in F2 values from AE participants, but not in F1 values, whereas their production of /ʌ/ differed significantly in both F1 and F2 values. This pattern mirrors Japanese speakers' performance with English /ɪ/-/ɪ/, where they distinguished /ɪ/-/R/ better than /ɪ/-/R/ because /ɪ/ is more similar to Japanese /R/. Accordingly, XNd participants' better production of /æ/ compared to /ʌ/ might be attributed to greater perceived phonetic distance of /æ/-/ɑ/ than /ʌ/-/ɑ/. Similarly, LSd participants' difficulty distinguishing the nonnative contrast /ʌ/-/ɑ/ might also stem from their inability to perceive the phonetic distance between the two sounds, as predicted by the SLM-r.

Another finding is from the durational data, where significant durational differences were found between XNd, LSd and AE speakers in the production of lax vowels, but not the tense ones. Specifically, both XNd and LSd groups produced /ɪ, ʌ, ɛ/ with significantly longer durations than AE participants. This can be explained by the Feature Hypothesis, which predicts that learners will have more difficulty acquiring L2 sounds that involve phonological features not present or not distinctive in their L1 (McAllister et al., 2002). The lack of lax-tense contrast in XNd, LSd and Mandarin likely contributed to these challenges. Nevertheless, XNd participants produced /ʊ/ with significantly longer durations than AE participants, while LSd participants showed no significant differences. The better performance of LSd participants on /ʊ/ might be attributed to the sound's presence in their dialect, further supporting SLM and PAM-L2 regarding the impact of L1 phonetic categories on L2 sound learning.

Nevertheless, although /ɛ/ occurs in the XNd, XNd participants had significantly longer durations in the production of /ɛ/ than AE participants. Interestingly, the F1 and F2 values of the XNd participants in producing the sound were not significantly different from those of AE participants. According to the PAM-L2, language learners assimilate an L2 sound into the most articulatory-similar sound in their L1 phonetic inventory (Best & Tyler, 2007). Hence, it was speculated that XNd participants might have benefited from the articulatory similarities (or the sameness) of the XNd /ɛ/ and AE /ɛ/, which are reflected in terms of F1 and F2 values. However, due to the lack of the duration feature in the L1 and L1 dialects, XNd participants failed to capture the duration features of the AE /ɛ/.

In addition, XNd speakers' non-nasalized production of /ɑ/ and /ɔ/ was unexpected, as it was hypothesized they would nasalize these sounds due to the presence of nasal counterparts in XNd. Their non-nasalized production of /ɔ/ might

be explained by its existence in XNd. In contrast, as /ɑ/ is absent in XNd and Mandarin, XNd participants likely assimilated it to the nearest articulatory counterpart /ã/, as suggested by PAM-L2. Their non-nasalized production of /ɑ/ might be explained by Feature Hypothesis: although /ɑ/ lacks in XNd, the oral-nasal feature exists in XNd and Mandarin.

Furthermore, considering the learning sequence perspective (XNd/LSd → Mandarin → English), XNd/LSd may serve as participants' L1, Mandarin as L2, and English as L3. Although this study does not focus on L2's impact on L3 learning, prior research addressing this issue suggests that L3 learners can leverage prior linguistic knowledge and experience to learn an additional language (Gut, 2010). Hence, it is plausible that XNd participants' success in distinguishing /ɑ/-/ã/ and /ɔ/-/õ/ benefited from their prior exposure to Mandarin, which lack nasal vowels.

In addition, both SLM and PAM-L2 suggest that new L2 categories can develop with increased L2 experience. Although all XNd and LSd participants had studied L2-English for over 10 years and possessed advanced proficiency, they still struggled to accurately produce all target vowels. This study only focused on their production performance, leaving unclear whether XNd participants might perform differently in perceiving tense-lax vowels.

8. Conclusion

This study compared the F₁, F₂ and duration values of 9 English vowels — /i/, /u/, /ɛ/, /ɔ/, /ɪ/, /æ/, /ʊ/, /ʌ/, and /ɑ/ — produced by XNd, LSd, and AE participants. Overall findings indicate that XNd and LSd participants had difficulties producing vowels absent from their dialects and Mandarin, aligning with the hypotheses of PAM-L2 and SLM. However, exceptions existed, such as XNd participants' significantly longer duration of /ɛ/ compared to AE participants, despite its presence in XNd. This discrepancy is likely due to the absence of a duration feature in Mandarin and XNd, supporting the Feature Hypothesis (McAllister et al., 2002). This is further evidenced by difficulties XNd and LSd participants faced distinguishing the lax-tense vowel pair /ɪ/-/i/. Moreover, despite many years of study and advanced English proficiency, challenges in producing certain AE vowels suggest that the influence of their L1 dialects persists, even with increased proficiency, contrary to what PAM-L2 and SLM predict.

These findings have important implications for L2 pronunciation instruction. When teaching L2-English pronunciation, for example, instructors should consider learner's L1 and L1 dialectal backgrounds to develop effective teaching materials and methods. Comparing the phonetic systems of English with those of

learners' L1 and dialects can help identify "problematic" sounds. Moreover, targeted teaching strategies can be implemented to address these challenges. Some misproduction may cause more difficulties in L2 intelligibility and comprehensibility than others (Munro & Derwing, 1995). For instance, emphasizing the distinction between /ɪ/ and /i/ is crucial due to the greater number of minimal pairs compared to /ʊ/ and /u/. Recognizing that L2 speakers from varying dialects encounter unique difficulties, teachers should allocate more time to practicing critical contrasts that significantly affect communication, thereby enhancing overall intelligibility and comprehensibility.

This study also has several limitations. First, more varied findings might emerge with participants having different English proficiency levels or diverse L1 dialects. Second, the focus was solely on AE vowels, and the number of AE participants was limited due to recruitment challenges among those matching the age range of XNd and LSd participants. Comparing outputs from a broader range of dialectal speakers with native speakers of a target L2 would be valuable. These factors will be considered in future research.

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Address for correspondence

Ying Li
Yangzhou University
China

liying_22@163.com



<https://orcid.org/0000-0003-1783-9083>

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